Customer Segmentation and Predictive Analytics System

Mini Project Report | Python, Streamlit, Pandas, scikit-learn and MySQL

1. Abstract

This project analyzes anonymized customer transaction data to understand purchasing behavior. It calculates Recency, Frequency and Monetary (RFM) features, assigns transparent rule-based customer segments, optionally compares them with K-means clusters, and demonstrates prediction of future 90-day spending from historical features. A Streamlit dashboard presents the results and allows CSV downloads.

2. Problem Statement

Retailers may have many transactions but limited visibility into which customers buy recently, purchase repeatedly, or contribute more revenue. This project turns transaction-level records into interpretable customer summaries that can support exploratory analysis and planning.

3. Objectives

Prepare anonymized data; calculate RFM features; create readable segment labels; visualize customer counts and spending; explore K-means clustering with scaled features; demonstrate a future-spending regression workflow; and document assumptions and limitations.

4. Tools and Technologies

Python; Streamlit; Pandas and NumPy for data preparation; Matplotlib for charts; scikit-learn for StandardScaler, K-means, silhouette score and RandomForestRegressor; MySQL for the relational schema; CSV for portable demo input.

5. Dataset Design

The supplied data is synthetic, not collected from real people. customers.csv contains customer_id, customer_code and optional signup_date. orders.csv contains order_id, customer_id, order_date and order_status. order_items.csv contains item_id, order_id, quantity and unit_price. Customer IDs are pseudonymous and do not identify actual people.

6. Methodology: RFM

Recency is the number of days between the analysis reference date and the most recent completed order. Frequency is the number of distinct completed orders. Monetary value is the sum of quantity multiplied by unit price for completed orders. Cancelled and returned orders are excluded. Rule-based labels use understandable thresholds, including Champions, Loyal / Active, At Risk, One-time, Regular and No Completed Purchase.

7. K-means Method

K-means is optional and groups customers using recency, frequency and monetary value. Frequency and monetary value are transformed using log1p to reduce skew, then all three features are standardized so their units do not dominate distance calculations. The user can choose K from 2 to 8. A silhouette score is reported as a rough cluster-separation indicator. It does not establish business usefulness, and numeric cluster IDs have no inherent meaning.

8. Predictive Analytics

The dashboard builds historical features from transactions before a cutoff and uses spend in the following 90-day period as the target. A Random Forest regressor estimates future spending. MAE and RMSE measure prediction error in currency units; R-squared is also displayed when meaningful. Because the dataset is synthetic and small, results are demonstrative only. A production project should use rolling temporal validation and compare against simple baselines.

9. Outputs

The dashboard displays customer and order counts, completed spending, RFM segment distribution, average spending by segment, segment profiles, customer-level RFM records, optional K-means profiles, prediction estimates, and downloadable CSV outputs.

10. Database Design

The SQL schema defines customers, orders, order_items, segment_runs and customer_segments. Foreign keys preserve relationships and indexes support customer/date queries. The current app reads CSVs so it runs without database credentials; the schema provides a foundation for a future MySQL integration.

11. Limitations and Ethics

Segments depend on chosen thresholds, data quality, analysis date and business context. They are not objective truths about customer intent. Small datasets can produce unstable clusters and misleading predictive metrics. Do not upload names, phone numbers, email addresses, payment details or other direct identifiers. Validate all results before making business decisions.

12. Conclusion

The project demonstrates an end-to-end customer analytics workflow from transaction preparation to segmentation, visualization and a basic spending prediction. It is suitable as an educational starting point and can be extended with a live MySQL connection, improved temporal validation, and domain-reviewed segment rules.

Example segment definitions

Segment	Illustrative meaning
Champions	Recent, frequent, high-spending under selected rules
Loyal / Active	Recent repeat purchasers
At Risk	Repeat purchasers whose last purchase was a long time ago
One-time	One completed order
Regular	Does not match another rule
No Completed Purchase	No completed order in the supplied data